Menghao Deng

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EDUCATION

The University of Hong Kong (HKU) 09/2025 –09/2029
- Doctor of Philosophy (Phd) Degree, Finance Hong Kong, China

The University of Hong Kong (HKU) 09/2021 – 06/2025
- Bachelor of Science (BSc) Degree, Major in **Mathematics** Hong Kong, China
- **Dean's Honours List** for 2023 – 2024 2024 –2025 Academic Year
- Computer Skills: Python (3yrs.), C/C++ (2yrs.), MATLAB (2yrs.), Linux shell (2yrs.), WIND (1yr.)

PUBLICATIONS & MANUSCRIPT

- Deng, M. (2024). The Analysis of Hidden Units in LSTM Model for Accurate Stock Price Prediction. In *Proceedings of the 1st International Conference on Data Analysis and Machine Learning – Volume 1: DAML*; ISBN 978-989-758-705-4, SciTePress, pp. 411-416. DOI: 10.5220/0012799100003885
- Deng, M. (2023). The Anatomy of Global Financial Crisis: A Comprehensive Analysis. *Advances in Economics, Management and Political Sciences*, 25, pp. 62-67. DOI: 10.54254/2754-1169/25/20230477
- Inference Method in LASSO Method. (in preparation)

INTERNSHIP EXPERIENCE

Haitong Securities Co., Ltd. 05/2023 – 09/2023
Intern, Bond Financing Department Shanghai, China
- Drafted five documents, including CMBS project approval application, REITs project approval application, CMBS resale and put plan, asset-backed plan application, and free trade bond issuance plan.
- Helped with due diligence by visiting clients, examining business operations, verifying information, and conducting risk screening.
- Supported two projects (with a scale of 30 billion RMB in total) by analyzing financial data, drafting the PPT of project proposals, engaging in five roadshows, scheduling the issuance timeline, responding to investor inquiries, writing social media posts, etc.
- Researched and designed new bonds to reduce financing costs through the bank guarantee model for the client, wrote bond issuance proposals and prospectus, and successfully implemented the Shanghai Stock Exchange's first guarantee letter model ABS.

Guotai Junan Securities 12/2022 – 02/2023
Intern, Investment Banking Department Remote
- Assisted in analyzing IPO data and failure cases (involving industries like healthcare and AI) from CITIC, CICC, and CITIC Construction Investment in the past three years; Utilized platforms like Snowball and Bloomberg to gather relevant information on cases (e.g., company profiles, financial data, shareholding penetration, etc.); Generated a 20-page report to enhance IPO risk control.
- Conducted industry research on wholesale and IT sectors by gathering and analyzing financial data (profit margins, debt ratios, etc.) of 27 leading companies to identify potential investment opportunities; Helped compile three thematic industry reports, totaling about 100,000 words.
- Contributed to eight IPO projects (with a scale of about 240 billion RMB in total) by drafting contract-based fund trust and asset management schemes during the listing period on the New Third Board and systematically analyzing about 60 potential enterprises, including public opinion monitoring, risk control, and competitor analysis.

RESEARCH EXPERIENCE

Centre of Quantitative History, HKU Business School 10/2023 – 08/2024
Research Assistant, Supervisor: Prof. Chen Lin Hong Kong, China
- Participated in a Python-based project to develop new arbitrage policies for low retracement and risk applicable to quantitative trading scenarios such as the gold and futures markets.
- Collected over 500 million A-share market announcements in China from 2020 to 2024 and conducted sentiment analysis using the Qwen-42 Large Language Model (LLM) to gain insights into market sentiment.
- Used the multiprocessing library to implement parallel processing for processing time reduction, and designed a

viable alternative approach (data imputation and bias-free smoothing) to handle unavailable announcements.

- Contributed to the design of the unified arbitrage policies by applying the Kalman Filtering for Regression to further standardize the processing of market data and integrating the results of sentiment analysis with the quantitative trading strategies.
- Helped with designing and debugging a back-testing framework to validate the viability of the arbitrage policies.

Guangdong Key Laboratory of Urban Informatics, Shenzhen University 06/2022 – 08/2022

Research Assistant, Supervisor: Prof. Dejin Zhang Shenzhen, China

Project: Learning Application and Improvement of Active Contour Model

- Gathered over 50 relevant studies for literature review, and helped formulate a research project proposal.
- Utilized Python to develop an adaptive intersection recognition program to advance the overall progress of image recognition.
- Implemented convolutional neural network to design and optimize an image recognition model, achieving image or video recognition in a manner that simulated human associative thinking.

PRACTICAL PROJECTS

Intensive Course on Investments (a seminar-based online program) 12/2022 – 02/2023

Participant, led by Professor Alexei Chekhlov from Columbia University Remote

- Obtained knowledge regarding investment management, quantitative finance, and stock investment, and became proficient in various indicator functions, investment analysis, portfolio design, etc.
- Worked with team members to construct and compare optimal portfolios using the Markowitz Model and Index Model by employing Python to process and analyze the monthly stock data of 10 companies across technology, energy, and consumer sectors from 2001 to 2021.

Machine Learning & Data Science: Application Development (a seminar-based program) 07/2023 – 08/2023

Participant, led by Professor Mark Vogelsberger from Massachusetts Institute of Technology Shanghai, China

- Mastered the application of deep learning algorithms based on sequence models such as LSTM and generative models such as GAN.
- Collaboratively completed a project by applying VGG16, GoogLeNet (Inception3), ResNet, and DenseNet121 to 12,000 images of drivers to recognize their behaviors. Performed data processing (data transformation and augmentation) and achieved high accuracy (over 95%) in all models except VGG16.

TEACHING EXPERIENCE

Department of Computer Science, HKU 09/2024 – Present

Student Teaching Assistant Hong Kong, China

- Assisted the professor in teaching and grading coursework for COMP1117 Computer Programming.
- Led tutoring sessions and provided one-on-one guidance to about 20 students on assignments and projects.

EXTRACURRICULAR ACTIVITIES

24 Frames Movie Club 09/2022 – 01/2023

Director & Screenwriter Hong Kong, China

- Organized and filmed five short films, one of which was presented at an on-campus film festival.

Game Chemistry (a game development studio) 10/2021 – 06/2022

Copywriter & Numerical Planning Shenzhen, China

- Operated a roguelike game on Taptap and wrote promotion articles. The downloads of the game reached 1 million.
- Worked on numerical models and formula design based on the game system rules.
- Created, configured, tested, and adjusted numerical-related tables to ensure game balance and scalability.

Online Contracted Writer (Qidian.com) 09/2021 – 01/2022

Contracted Writer Remote

- Finished a long fiction of more than 300,000 Chinese characters and was invited to be a contracted writer.

OTHER INFORMATION

- **The 3rd Place** in 2022 Bank of Communication & Hong Kong Nine Universities E-sports League 03/2023
- **Hobbies:** creative writing, football, swimming, badminton, drum kit (passed the amateur grade 9), guitar